Filter Summary Report: CG,TIA,Automated,simple,Z2,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z2 ZL: $\frac{Z_2 Z_L g_m + Z_L}{Z_2 g_m + 1}$

$$H(z) = \frac{Z_2 Z_L g_m + Z_L}{Z_2 g_m + 1}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.2 BP-2 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.3 BP-3 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s + 1},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.5 BP-5 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.6 BP-6 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

3.7 BP-7 $Z(s) = \left(\infty,\ \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1},\ \infty,\ \infty,\ \infty,\ \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}\right)$

$$H(s) = \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_L R_L}$
K-LP: 0
K-HP: 0

K-BP: R_L
Qz: None
Wz: None

3.8 BP-8 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{L_LR_Ls}{C_LL_LR_Ls^2+L_Ls+R_L}\right)$

$$H(s) = \frac{L_LR_Ls}{C_LL_LR_Ls^2 + L_Ls + R_L}$$

Parameters:

Q: $\frac{\sqrt{C_L}R_L}{\sqrt{L_L}}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{1}{C_LR_L}$
K-LP: 0
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.2 BS-2 $Z(s) = \left(\infty,\ \frac{1}{C_2s},\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.3 BS-3 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.4 BS-4 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.5 BS-5 $Z(s) = \left(\infty,\ L_2s + \frac{1}{C_2s},\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.6 BS-6 $Z(s) = \left(\infty,\ L_2s + R_2 + \frac{1}{C_2s},\ \infty,\ \infty,\ \infty,\ \frac{R_L\left(C_LL_Ls^2+1\right)}{C_LL_Ls^2+C_LR_Ls+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + R_L}{C_LL_Ls^2 + C_LR_Ls + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L}R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$

K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.7 BS-7 $Z(s) = \left(\infty,\ \frac{C_2 L_2 R_2 s^2 + L_2 s + R_2}{C_2 L_2 s^2 + 1},\ \infty,\ \infty,\ \infty,\ \frac{R_L \left(C_L L_L s^2 + 1\right)}{C_L L_L s^2 + C_L R_L s + 1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + R_L}{C_L L_L s^2 + C_L R_L s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L} R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.8 BS-8 $Z(s) = \left(\infty,\ \frac{R_2 \left(C_2 L_2 s^2 + 1\right)}{C_2 L_2 s^2 + C_2 R_2 s + 1},\ \infty,\ \infty,\ \infty,\ \frac{R_L \left(C_L L_L s^2 + 1\right)}{C_L L_L s^2 + C_L R_L s + 1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + R_L}{C_L L_L s^2 + C_L R_L s + 1}$$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L} R_L}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{R_L}{L_L}$
K-LP: R_L
K-HP: R_L
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ R_L)$

$$H(s) = R_L$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_L R_2 R_L g_m + C_L R_L\right) + 1}{s\left(C_L R_2 g_m + C_L\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_2 g_m + s^2\left(C_L L_L R_2 g_m + C_L L_L\right) + 1}{s\left(C_L R_2 g_m + C_L\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_2 g_m + s^2\left(C_L L_L R_2 g_m + C_L L_L\right) + s\left(C_L R_2 R_L g_m + C_L R_L\right) + 1}{s\left(C_L R_2 g_m + C_L\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ R_2,\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L R_L s^2 + g_m + s\left(C_2 + C_L R_L g_m\right)}{C_2 C_L s^2 + C_L g_m s}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_L s^3 + C_2 s + C_L L_L g_m s^2 + g_m}{C_2 C_L s^2 + C_L g_m s}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_L s^3 + g_m + s^2\left(C_2 C_L R_L + C_L L_L g_m\right) + s\left(C_2 + C_L R_L g_m\right)}{C_2 C_L s^2 + C_L g_m s}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L R_2 R_L s^2 + R_2 g_m + s\left(C_2 R_2 + C_L R_2 R_L g_m + C_L R_L\right) + 1}{C_2 C_L R_2 s^2 + s\left(C_L R_2 g_m + C_L\right)}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_L R_2 s^3 + C_2 R_2 s + R_2 g_m + s^2 \left(C_L L_L R_2 g_m + C_L L_L\right) + 1}{C_2 C_L R_2 s^2 + s \left(C_L R_2 g_m + C_L\right)}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_L R_2 s^3 + R_2 g_m + s^2 \left(C_2 C_L R_2 R_L + C_L L_L R_2 g_m + C_L L_L\right) + s \left(C_2 R_2 + C_L R_2 R_L g_m + C_L R_L\right) + 1}{C_2 C_L R_2 s^2 + s \left(C_L R_2 g_m + C_L\right)}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m + s^2 \left(C_2 C_L R_2 R_L g_m + C_2 C_L R_L\right) + s \left(C_2 R_2 g_m + C_2 + C_L R_L g_m\right)}{C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L L_L g_m s^2 + g_m + s^3 \left(C_2 C_L L_L R_2 g_m + C_2 C_L L_L\right) + s \left(C_2 R_2 g_m + C_2\right)}{C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m + s^3 \left(C_2 C_L L_L R_2 g_m + C_2 C_L L_L\right) + s^2 \left(C_2 C_L R_2 R_L g_m + C_2 C_L R_L + C_L L_L g_m\right) + s \left(C_2 R_2 g_m + C_2 + C_L R_L g_m\right)}{C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 R_L g_m s^3 + g_m + s^2 \left(C_2 C_L R_L + C_2 L_2 g_m\right) + s \left(C_2 + C_L R_L g_m\right)}{C_2 C_L L_2 g_m s^3 + C_2 C_L s^2 + C_L g_m s}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 L_L g_m s^4 + C_2 C_L L_L s^3 + C_2 s + g_m + s^2 \left(C_2 L_2 g_m + C_L L_L g_m\right)}{C_2 C_L L_2 g_m s^3 + C_2 C_L s^2 + C_L g_m s}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 L_L g_m s^4 + g_m + s^3 \left(C_2 C_L L_2 R_L g_m + C_2 C_L L_L\right) + s^2 \left(C_2 C_L R_L + C_2 L_2 g_m + C_L L_L g_m\right) + s \left(C_2 + C_L R_L g_m\right)}{C_2 C_L L_2 g_m s^3 + C_2 C_L s^2 + C_L g_m s}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(\infty,\ L_2 s + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 R_L g_m s^3 + g_m + s^2 \left(C_2 C_L R_2 R_L g_m + C_2 C_L R_L + C_2 L_2 g_m\right) + s \left(C_2 R_2 g_m + C_2 + C_L R_L g_m\right)}{C_2 C_L L_2 g_m s^3 + C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 L_L g_m s^4 + g_m + s^3 \left(C_2 C_L L_L R_2 g_m + C_2 C_L L_L\right) + s^2 \left(C_2 L_2 g_m + C_L L_L g_m\right) + s \left(C_2 R_2 g_m + C_2\right)}{C_2 C_L L_2 g_m s^3 + C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{L_L s}{C_L L_L s^2 + 1}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_2 C_L L_2 L_L g_m s^4 + g_m + s^3 \left(C_2 C_L L_2 R_L g_m + C_2 C_L L_L R_2 g_m + C_2 C_L L_L\right) + s^2 \left(C_2 C_L R_2 R_L g_m + C_2 C_L R_L + C_2 L_2 g_m + C_L L_L g_m\right) + s \left(C_2 R_2 g_m + C_2 + C_L R_L g_m\right)}{C_2 C_L L_2 g_m s^3 + C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L\right)}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\infty,\ L_2 s + R_2 + \frac{1}{C_2 s},\ \infty,\ \infty,\ \infty,\ \frac{C_L L_L R_L s^2+L_L s+R_L}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\infty,\ \frac{C_2 L_2 R_2 s^2+L_2 s+R_2}{C_2 L_2 s^2+1},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\infty,\ \frac{C_2 L_2 R_2 s^2+L_2 s+R_2}{C_2 L_2 s^2+1},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{C_L s}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_L}{C_LR_Ls+1}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ R_L+\frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_2g_m + s^3\left(C_2C_LL_2R_2R_Lg_m + C_2C_LL_2R_L\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_LL_2R_Lg_m\right) + s\left(C_LR_2R_Lg_m + C_LR_L + L_2g_m\right) + 1}{C_LL_2g_ms^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ L_Ls+\frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_LL_2L_Lg_ms^3 + L_2g_ms + R_2g_m + s^4\left(C_2C_LL_2L_LR_2g_m + C_2C_LL_2L_L\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_LL_LR_2g_m + C_LL_L\right) + 1}{C_LL_2g_ms^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{L_Ls}{C_LL_Ls^2+1}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ L_Ls+R_L+\frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_LL_2L_LR_2g_m + C_2C_LL_2L_L\right) + s^3\left(C_2C_LL_2R_2R_Lg_m + C_2C_LL_2R_L + C_LL_2L_Lg_m\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_LL_2R_Lg_m + C_LL_LR_2g_m + C_LL_L\right) + s\left(C_LR_2R_Lg_m + C_LR_L + L_2g_m\right) + 1}{C_LL_2g_ms^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\infty,\ \frac{C_2L_2R_2s^2+L_2s+R_2}{C_2L_2s^2+1},\ \infty,\ \infty,\ \infty,\ \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1}$$

9.57 X-INVALID-ORDER-57 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = R_L$$

9.58 X-INVALID-ORDER-58 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{1}{C_Ls}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_L}{C_LR_Ls+1}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_2g_m + s^3\left(C_2C_LL_2R_2R_Lg_m + C_2C_LL_2R_L\right) + s^2\left(C_2C_LR_2R_L + C_2L_2R_2g_m + C_2L_2\right) + s\left(C_2R_2 + C_LR_2R_Lg_m + C_LR_L\right) + 1}{C_2C_LR_2s^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ L_Ls + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{C_2C_LL_LR_2s^3 + C_2R_2s + R_2g_m + s^4\left(C_2C_LL_2L_LR_2g_m + C_2C_LL_2L_L\right) + s^2\left(C_2L_2R_2g_m + C_2L_2 + C_LL_LR_2g_m + C_LL_L\right) + 1}{C_2C_LR_2s^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{L_Ls}{C_LL_Ls^2 + 1}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ L_Ls + R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_2g_m + s^4\left(C_2C_LL_2L_LR_2g_m + C_2C_LL_2L_L\right) + s^3\left(C_2C_LL_2R_2R_Lg_m + C_2C_LL_2R_L + C_2C_LL_LR_2\right) + s^2\left(C_2C_LR_2R_L + C_2L_2R_2g_m + C_2L_2 + C_LL_LR_2g_m + C_LL_L\right) + s\left(C_2R_2 + C_LR_2R_Lg_m + C_LR_L\right) + 1}{C_2C_LR_2s^2 + s^3\left(C_2C_LL_2R_2g_m + C_2C_LL_2\right) + s\left(C_LR_2g_m + C_L\right)}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(\infty,\ \frac{R_2\left(C_2L_2s^2+1\right)}{C_2L_2s^2+C_2R_2s+1},\ \infty,\ \infty,\ \infty,\ \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{C_LL_LR_Ls^2 + L_Ls + R_L}{C_LL_Ls^2 + 1}$$

10 X-INVALID-WZ

11 X-PolynomialError